

# Google Cloud Deployment Guide — Nucleus AI

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Deploy Nucleus AI on GCP using **Cloud Run** (serverless) or **GKE** (Kubernetes).

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## Prerequisites

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- Google Cloud SDK ( `gcloud` ) installed and authenticated
  - Docker installed locally
  - A GCP project with billing enabled
  - Domain name configured
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## Option A: Cloud Run (Recommended for Simplicity)

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### 1. Enable Required APIs

```
gcloud services enable \
  run.googleapis.com \
  artifactregistry.googleapis.com \
  sqladmin.googleapis.com \
  secretmanager.googleapis.com \
  cloudbuild.googleapis.com
```

### 2. Create Artifact Registry

```
gcloud artifacts repositories create nucleus-ai \
  --repository-format=docker \
  --location=us-central1 \
  --description="Nucleus AI container images"

# Authenticate Docker
gcloud auth configure-docker us-central1-docker.pkg.dev
```

### 3. Build and Push Images

```
PROJECT_ID=$(gcloud config get-value project)
REGION=us-central1
REGISTRY=${REGION}-docker.pkg.dev/${PROJECT_ID}/nucleus-ai

# Build and push backend
docker build -f docker/production/Dockerfile.backend -t ${REGISTRY}/backend:latest .
docker push ${REGISTRY}/backend:latest

# Build and push frontend
docker build -f docker/production/Dockerfile.frontend \
  --build-arg NEXT_PUBLIC_API_URL=https://api.yourdomain.com \
  -t ${REGISTRY}/frontend:latest .
docker push ${REGISTRY}/frontend:latest
```

### 4. Set Up Cloud SQL (PostgreSQL)

```
# Create instance
gcloud sql instances create nucleus-db \
  --database-version=POSTGRES_16 \
  --tier=db-custom-2-4096 \
  --region=us-central1 \
  --storage-type=SSD \
  --storage-size=20GB \
  --storage-auto-increase \
  --backup-start-time=03:00 \
  --availability-type=REGIONAL \
  --root-password="YOUR_STRONG_PASSWORD"

# Create database
gcloud sql databases create nucleus --instance=nucleus-db

# Create user
gcloud sql users create nucleus \
  --instance=nucleus-db \
  --password="YOUR_DB_PASSWORD"
```

### 5. Store Secrets

```
echo -n "sk-your-openai-key" | \
  gcloud secrets create OPENAI_API_KEY --data-file=-

echo -n "your-nextauth-secret" | \
  gcloud secrets create NEXTAUTH_SECRET --data-file=-

echo -n "your-secret-key" | \
  gcloud secrets create SECRET_KEY --data-file=-
```

## 6. Deploy Backend to Cloud Run

```
gcloud run deploy nucleus-backend \
  --image=${REGISTRY}/backend:latest \
  --region=us-central1 \
  --platform=managed \
  --port=8000 \
  --memory=2Gi \
  --cpu=2 \
  --min-instances=1 \
  --max-instances=10 \
  --timeout=300 \
  --concurrency=80 \
  --set-env-vars="APP_ENV=production,DEBUG=false,QDRANT_URL=https://your-
qdrant.cloud.qdrant.io" \
  --set-secrets="OPENAI_API_KEY=OPENAI_API_KEY:latest,SECRET_KEY=SECRET_KEY:latest" \
  --add-cloudsql-instances=${PROJECT_ID}:us-central1:nucleus-db \
  --set-env-vars="DATABASE_URL=postgresql+asyncpg://nucleus:PASS@nucleus?host=/cloud-
sql/${PROJECT_ID}:us-central1:nucleus-db" \
  --allow-unauthenticated
```

## 7. Deploy Frontend to Cloud Run

```
gcloud run deploy nucleus-frontend \
  --image=${REGISTRY}/frontend:latest \
  --region=us-central1 \
  --platform=managed \
  --port=3000 \
  --memory=1Gi \
  --cpu=1 \
  --min-instances=1 \
  --max-instances=5 \
  --timeout=60 \
  --set-env-vars="NODE_ENV=production,NEXTAUTH_URL=https://yourdomain.com" \
  --set-secrets="NEXTAUTH_SECRET=NEXTAUTH_SECRET:latest" \
  --allow-unauthenticated
```

## 8. Set Up Custom Domain

```
# Map domain to frontend service
gcloud run domain-mappings create \
  --service=nucleus-frontend \
  --domain=yourdomain.com \
  --region=us-central1

# Map API subdomain to backend
gcloud run domain-mappings create \
  --service=nucleus-backend \
  --domain=api.yourdomain.com \
  --region=us-central1
```

Cloud Run provides automatic SSL certificates via managed Google certificates.

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## Option B: GKE (Kubernetes)

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For teams needing full Kubernetes control.

### 1. Create GKE Cluster

```
gcloud container clusters create-auto nucleus-ai-cluster \
  --region=us-central1 \
  --release-channel=regular
```

## 2. Create Kubernetes Manifests

```
# k8s/backend-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nucleus-backend
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nucleus-backend
  template:
    metadata:
      labels:
        app: nucleus-backend
    spec:
      containers:
        - name: backend
          image: us-central1-docker.pkg.dev/PROJECT/nucleus-ai/backend:latest
          ports:
            - containerPort: 8000
          resources:
            requests:
              cpu: 500m
              memory: 512Mi
            limits:
              cpu: "2"
              memory: 2Gi
          livenessProbe:
            httpGet:
              path: /health
              port: 8000
            initialDelaySeconds: 15
            periodSeconds: 30
          readinessProbe:
            httpGet:
              path: /health
              port: 8000
            initialDelaySeconds: 10
            periodSeconds: 10
          envFrom:
            - secretRef:
                name: nucleus-secrets
---
apiVersion: v1
kind: Service
metadata:
  name: nucleus-backend
spec:
  selector:
    app: nucleus-backend
  ports:
    - port: 8000
      targetPort: 8000
  type: ClusterIP
```

### 3. Deploy

```
kubectl apply -f k8s/
```

## Monitoring

- **Cloud Monitoring:** Automatic metrics for Cloud Run (latency, request count, error rate)
- **Cloud Logging:** Structured logs from containers
- **Error Reporting:** Automatic error grouping and alerting
- **Uptime Checks:** Configure in Cloud Monitoring console

## Cost Estimation (Monthly)

| Service            | Spec                         | Estimated Cost   |
|--------------------|------------------------------|------------------|
| Cloud Run Backend  | 2 vCPU, 2 GB, min 1 instance | ~\$50            |
| Cloud Run Frontend | 1 vCPU, 1 GB, min 1 instance | ~\$25            |
| Cloud SQL          | db-custom-2-4096, HA         | ~\$80            |
| Qdrant Cloud       | Starter                      | ~\$25            |
| Artifact Registry  | Image storage                | ~\$5             |
| <b>Total</b>       |                              | <b>~\$185/mo</b> |